

08. Messhilite Inheritance Analysis

One-sentence summary

In the tested Rune Factory 5 Messhilite-inheritance conditions, increasing the number of copies of the same target material increased the number of successful three-slot combinations and reduced the expected number of retries; adding unrelated materials increased the candidate space without necessarily increasing successful combinations.

This chapter examines Messhilite inheritance in *Rune Factory 4 Special* (RF4SP) and *Rune Factory 5* (RF5). It combines a combination-space reference model with repeated in-game observations. It does not claim access to, or proof of, the games' internal implementation.

The principal RF5 validation contains 4,000 trials: 1,000 trials each for the 3-stack, 4-stack, 5-stack, and 6-stack conditions. RF4SP also includes a separate four-condition validation with 100 trials per condition (400 total), plus smaller five-trial exploratory comparisons. The 400-trial dataset supports a condition-specific quantitative comparison; the five-trial checks remain exploratory.

8-1. Research question and practical objective

Messhilite inheritance is often experienced as pure luck. The practical question is more specific:

Can a player increase the probability of retaining three copies of the intended arrange material while using Messhilite Ore for performance inheritance?

The important distinction is not simply how many materials are added, but **which materials are added**.

- Adding another copy of the same target material can increase both the total candidate count and the number of successful three-candidate subsets.
- Adding a different material can increase the total candidate count without increasing the number of subsets that satisfy the target condition.

For a common practical use case, the target material may be Glitter Augite for weapon range or Love Crystal for drain-attack testing. The logic applies only to conditions in which three copies of the same target material constitute success.

Confirmed observation: coexistence is possible

An RF5 observation confirmed that Messhilite inheritance and a three-slot arrange containing target materials can coexist. In one observed construction, the completed weapon retained two Glitter Augites together with Messhilite Ore in the arrange display. The precise displayed order was not preserved with sufficient confidence, so this observation establishes coexistence only; it does not establish an order rule.

8-2. Analytical model and its boundary

Let:

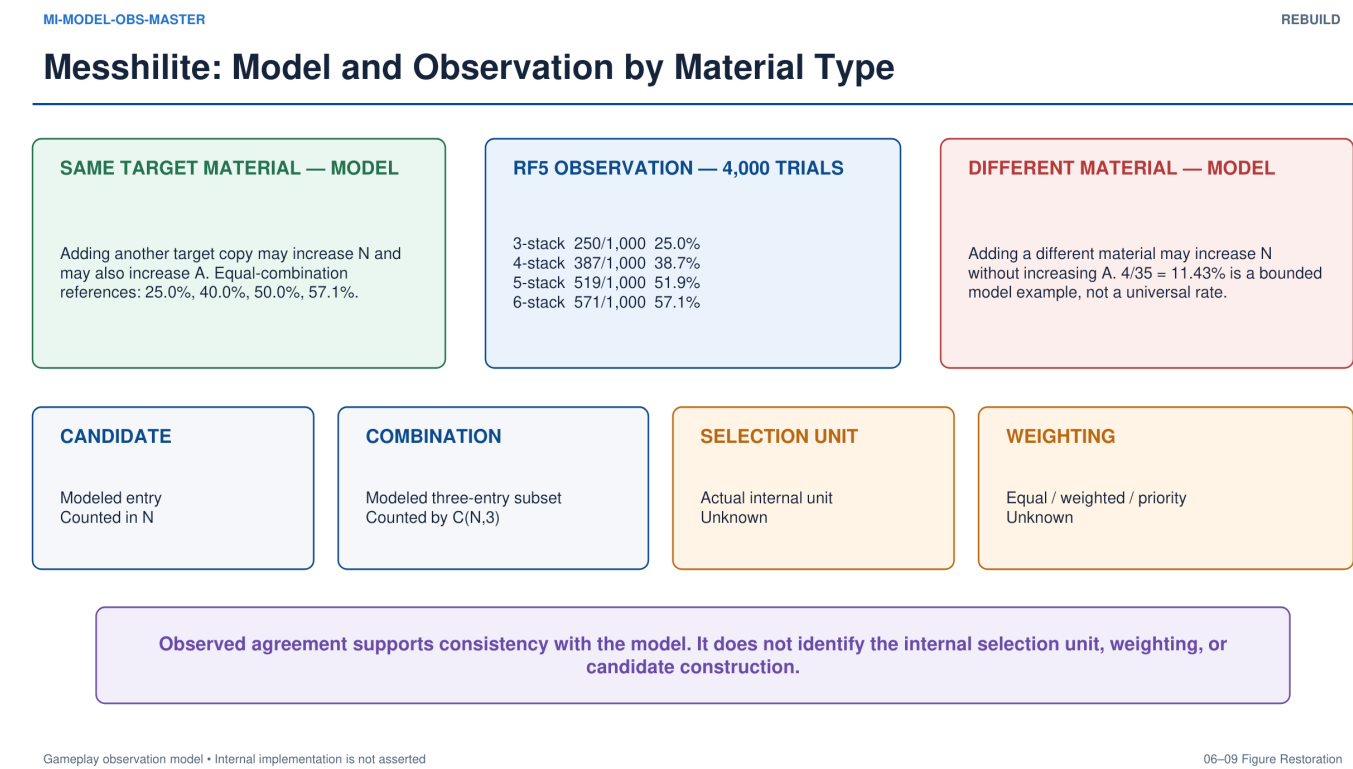
- $M = 3$: the number of retained arrange slots in the tested setup;
- N : the total number of candidates, including Messhilite Ore;
- T : the number of copies of the target material among those candidates;
- $K(N,M) = C(N,M)$: the size of the possible three-candidate combination space when $N > M$;
- $W = C(T,M)$: the number of three-candidate subsets composed entirely of the target material.

Under a **simple equal-selection approximation**, the reference probability is:

$$P_{\text{reference}} = W / K(N,M) = C(T,3) / C(N,3)$$

Figure 08-P02-img-000

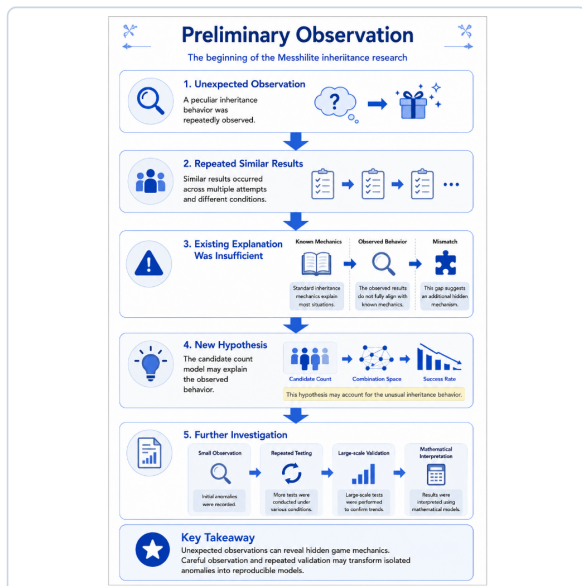
Master: MI-MODEL-OBS-MASTER | Source anchor: Japanese Canonical p.2



Master: MESSHILITE-PRELIMINARY-OBS | Source anchor: Japanese Canonical p.3

REUSE / TRANSLATE

Messhilite Preliminary Observation



EXPLANATORY FUNCTION RETAINED

The source figure retains the input materials, crafting record, displayed result, preliminary comparison, and follow-up question.

CURRENT INTERPRETATION

Preserve the observed input/output record. The preliminary difference motivates validation but does not identify the causal transition.

SOURCE DETAIL + CURRENT BOUNDARY

Gameplay observation model • Internal implementation is not asserted

06-09 Figure Restoration

Placement caption

Messhilite + Glitter observation strip

Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

This expression describes a combination-space approximation. It does not prove that the game selects every subset with equal probability. Chapter 06 showed a strongly non-uniform retained-subset distribution under a different RF4SP turnip-family condition, and Chapter 07 therefore limits equal selection to a conditional approximation.

The stack labels used in this chapter refer to the number of copies of the same target material, not to the total candidate count. Messhilite Ore itself also enters the candidate pool, so the candidate count is one larger.

Test condition	Target copies T	Total candidates N	Successful subsets C(T,3)	All subsets C(N,3)	Reference probability
3-stack	3	4	1	4	25.0%
4-stack	4	5	4	10	40.0%
5-stack	5	6	10	20	50.0%
6-stack	6	7	20	35	57.1%

The model's useful claim is limited: same-material stacking creates additional winning subsets. Whether the game distributes outcomes uniformly over those subsets remains an empirical question for each tested condition.

8-3. RF5 validation design

Environment and judgment rule

- Game: Rune Factory 5
- Platform: Steam
- Target mechanic: Messhilite inheritance
- Judgment material: Love Crystal
- Conditions: 3-stack, 4-stack, 5-stack, and 6-stack
- Trials: 1,000 per condition; 4,000 total

The completed weapon's drain-attack rate was used as the judgment interface:

Observed drain rate	Judgment
9%	Success: three Love Crystal effects retained
6%	Failure: one target-material effect was absent or replaced

The data were recorded in 100-trial blocks. Dropdown entry was used to reduce input mistakes, and success/failure was calculated automatically from the recorded drain rate. Aggregated data were then produced from the block-level records.

Why 1,000 trials per condition were used

Early 100-trial blocks fluctuated substantially. Individual blocks produced rates ranging roughly from the mid-30% range to around 60%, depending on condition and block. A 100-trial result was therefore considered too unstable for the intended comparison, and each condition was extended to 1,000 trials.

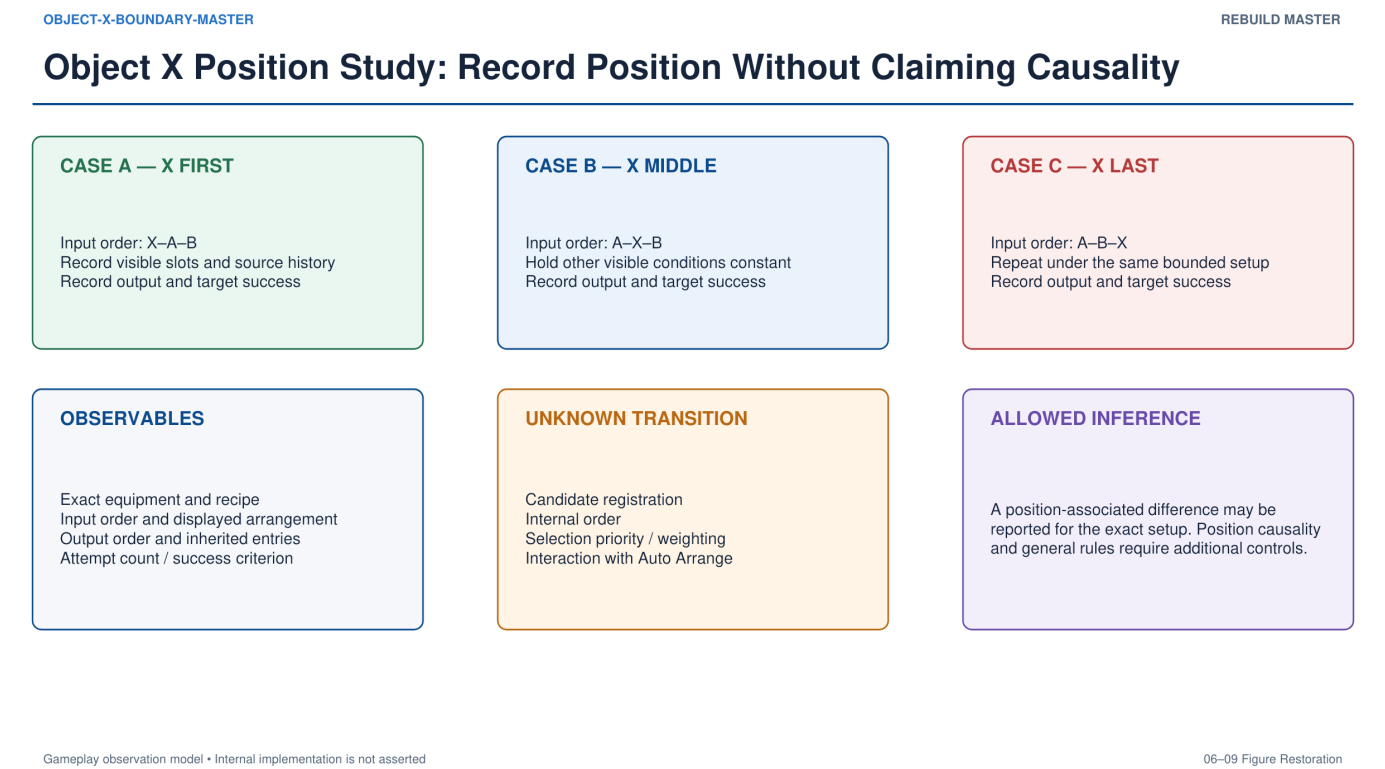
The published package retains the raw integrated CSV, condition-level summaries, the validation method, and the Wilson-interval derivation so that the aggregation route can be checked independently.

8-4. RF5 results

Condition	Trials	Successes	Observed rate	95% Wilson interval	Reference model	Difference
3-stack	1,000	250	25.0%	22.4%-27.8%	25.0%	0.0 pt
4-stack	1,000	387	38.7%	35.7%-41.8%	40.0%	-1.3 pt
5-stack	1,000	519	51.9%	48.8%-55.0%	50.0%	+1.9 pt
6-stack	1,000	571	57.1%	54.0%-60.1%	57.1%	0.0 pt

Figure 08-P05-img-002

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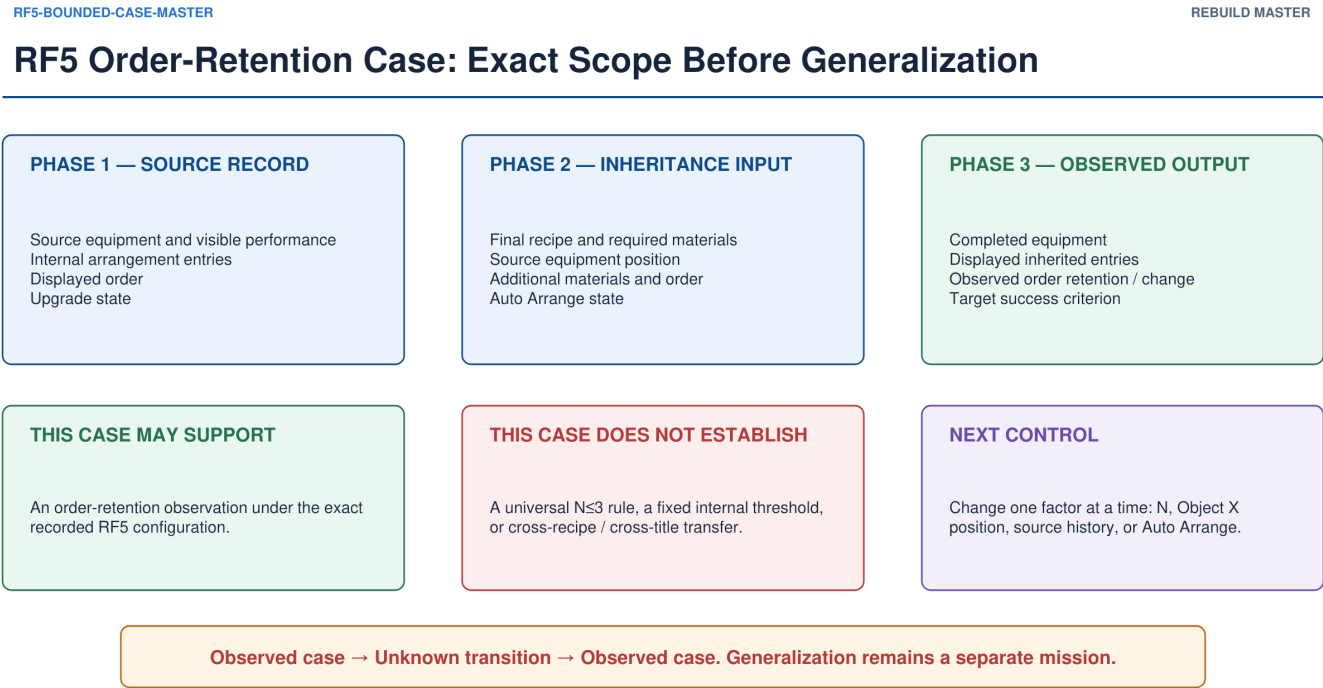


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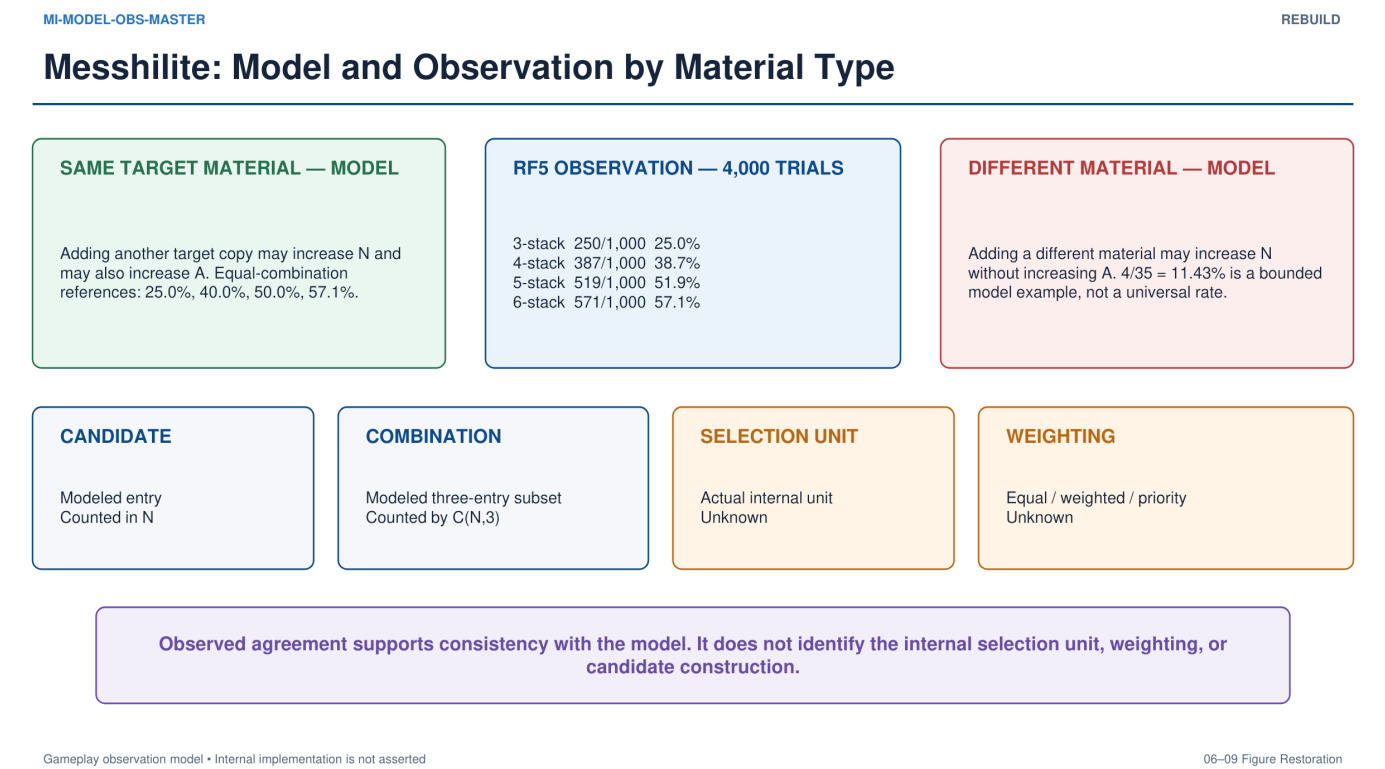
Object X position model
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Figure 08-P06-img-003

Master: RF5-BOUNDED-CASE-MASTER | Source anchor: Japanese Canonical p.6



Placement caption
RF5 $N \leq 3$ order-retention model
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.



All four reference values fall inside the corresponding 95% Wilson intervals. Under these RF5 test conditions, the observations are therefore consistent with the equal-selection combination approximation.

This is not evidence that every candidate-overflow condition in RF5 is uniform. It is evidence that this particular same-material Messhilite test series behaved close to the reference model at the available resolution.

Marginal improvement

Change	Observed improvement	Practical interpretation
3-stack -> 4-stack	+13.7 percentage points	First additional copy beyond the three target slots; strong gain per added material
4-stack -> 5-stack	+13.2 percentage points	Raises observed success above 50%; useful for stability
5-stack -> 6-stack	+5.2 percentage points	Highest observed rate, but diminishing return

The observed results support a practical decision rule:

- **Minimum-material option:** 3-stack.
- **Efficiency-oriented option:** 4-stack. It produces the first large gain beyond the three-slot baseline.
- **Stability-oriented option:** 5-stack. It reached 51.9% and reduced the expected retry burden to roughly two attempts per success.
- **Maximum tested rate:** 6-stack. It reached 57.1%, but the improvement over 5-stack was much smaller.

These are practical recommendations derived from the tested conditions, not universal optimization rules for every recipe or material.

8-5. Same-material stacking versus different-material mixing

The same total candidate count can have very different practical meaning depending on the composition of the candidate pool.

Complete same-material target

When success requires three copies of the same target material, adding another copy increases the number of winning subsets:

$$C(3,3) = 1, C(4,3) = 4, C(5,3) = 10, C(6,3) = 20.$$

This is why the observed 3-stack through 6-stack rates rise rather than fall.

Three different target materials

If success instead requires one copy each of three different materials, repeatedly adding only one of them may enlarge the candidate pool without adding enough new winning structures. A representative equal-selection reference falls from 25.0% at four candidates to approximately 11.4% at seven candidates.

This 57.1% -> 11.4% contrast is not caused by candidate count alone. It arises because the numerator - the number of successful candidate subsets - changes differently under the two material compositions.

Mixed targets

Two-of-one-material plus one-of-another conditions lie between the complete same-material and complete different-material cases. Their probabilities depend on which material is duplicated, which material is

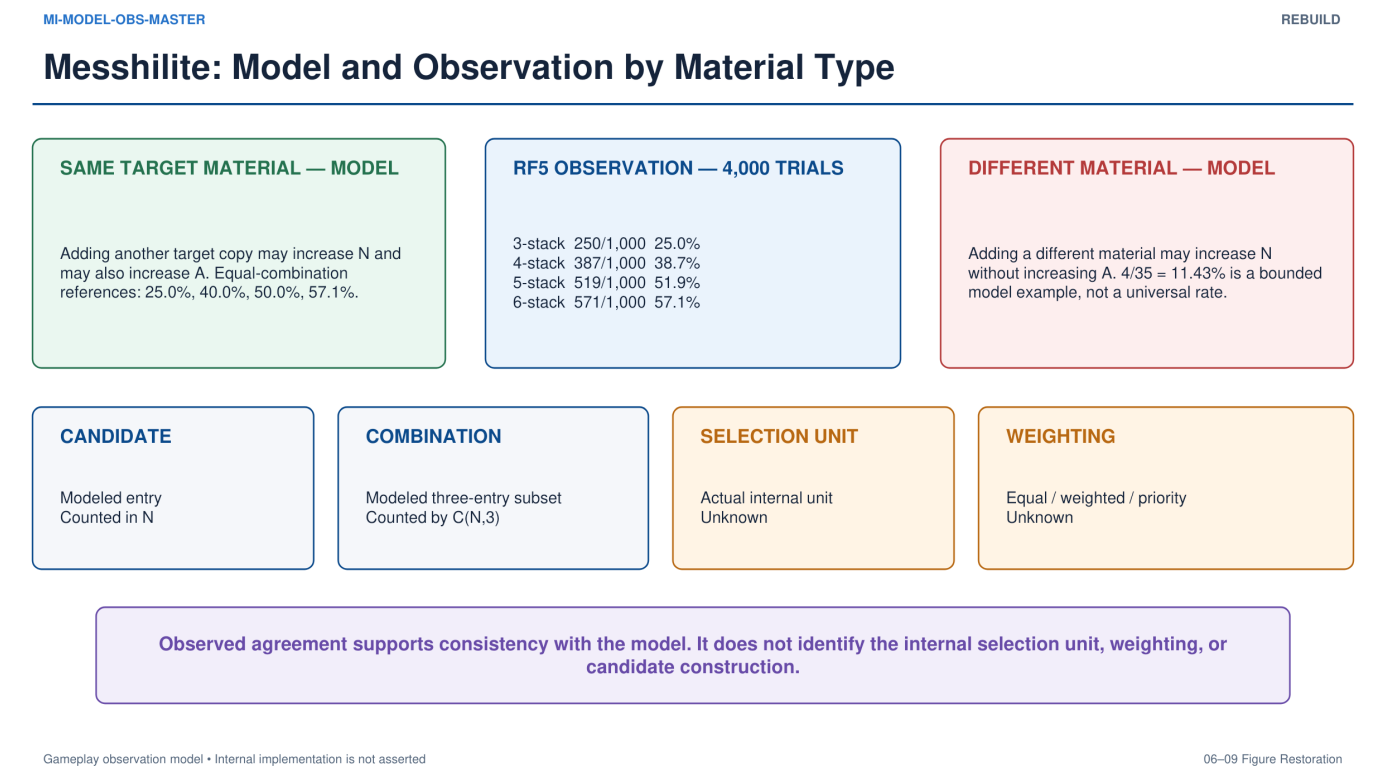


Figure 08-P09-img-006

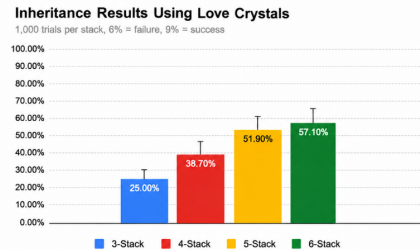
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MESSHILITE-4000-OBS

REUSE / TRANSLATE

RF5 Messhilite Validation: 4,000 Trials

	3-Stack	4-Stack	5-Stack	6-Stack
Success Rate	25.00%	38.70%	51.90%	57.10%
95% Confidence Interval (Lower Bound)	22.32%	35.68%	48.80%	54.03%
95% Confidence Interval (Upper Bound)	27.68%	41.72%	55.00%	60.17%
95% Wilson Confidence Interval (Lower Bound)	22.42%	35.73%	48.80%	54.01%
95% Wilson Confidence Interval (Upper Bound)	27.78%	41.76%	54.98%	60.13%
Number of Trials	1,000	1,000	1,000	1,000
Number of Successes	250	387	519	571
Stack Ratio (vs 3-Stack)	1.00	1.55	2.08	2.28



EXPLANATORY FUNCTION RETAINED

The source figure retains stack conditions, sample sizes, observed rates, confidence-interval context, and the Love Crystal comparison.

CURRENT INTERPRETATION

Observed successes per 1,000 trials are 250, 387, 519, and 571. Agreement with a reference model does not prove implementation.

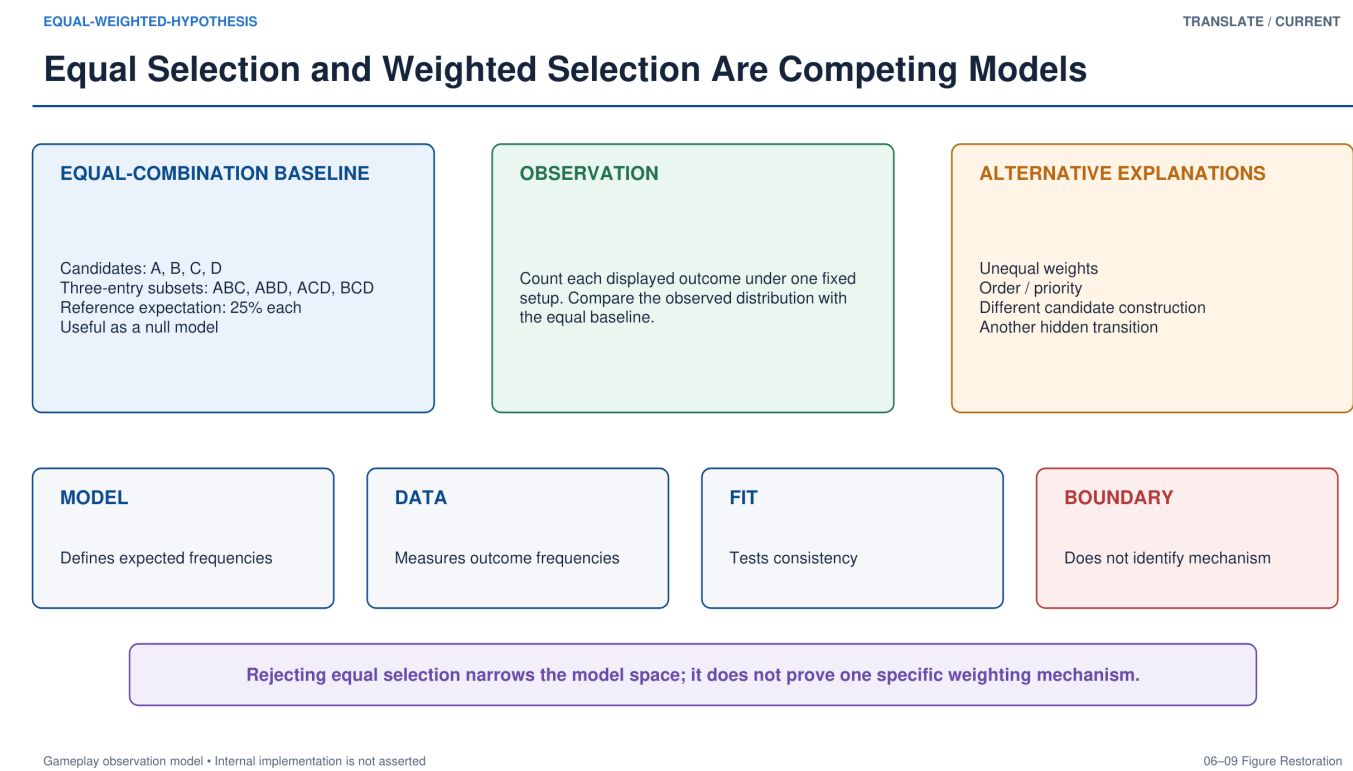
SOURCE DETAIL + CURRENT BOUNDARY

Gameplay observation model • Internal implementation is not asserted

06–09 Figure Restoration

Placement caption

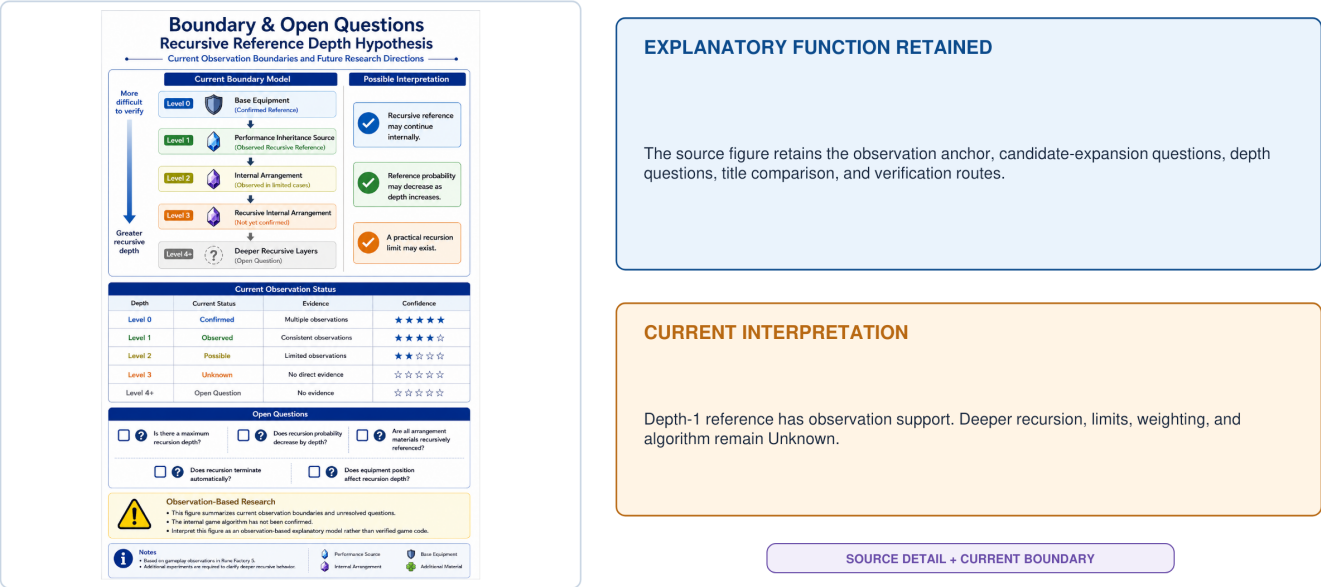
60-case Messhilite success-rate panel
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.



Placement caption

Equal versus weighted selection model
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Recursive Processing: Boundary and Open Questions



added, and what exact subset constitutes success. They should be calculated from the stated success condition rather than assigned a generic rate.

The broader rule is:

Candidate count organizes the size of the combination space. Material composition determines how many positions in that space satisfy the target condition. The probability distribution over the space must still be observed.

8-6. RF4SP boundary and cross-title interpretation

RF4SP 400-trial validation

A separate RF4SP dataset was collected after the permutation problem identified in Chapter 07. It had two purposes: to test whether the material-condition-dependent order bias had also broken the Candidate Count Model as a combination-space reference, and to test whether matched stack conditions produced detectably different success rates for Glitter Augite and Love Crystal. Four same-material Messhilite-inheritance conditions were observed, with 100 trials per condition and 400 trials in total.

Condition	Trials	Successes	Observed rate	Reference rate
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Love Crystal 3-stack	100	29	29%	25.00%
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Glitter Augite 3-stack	100	28	28%	25.00%
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Love Crystal 6-stack	100	59	59%	57.14%
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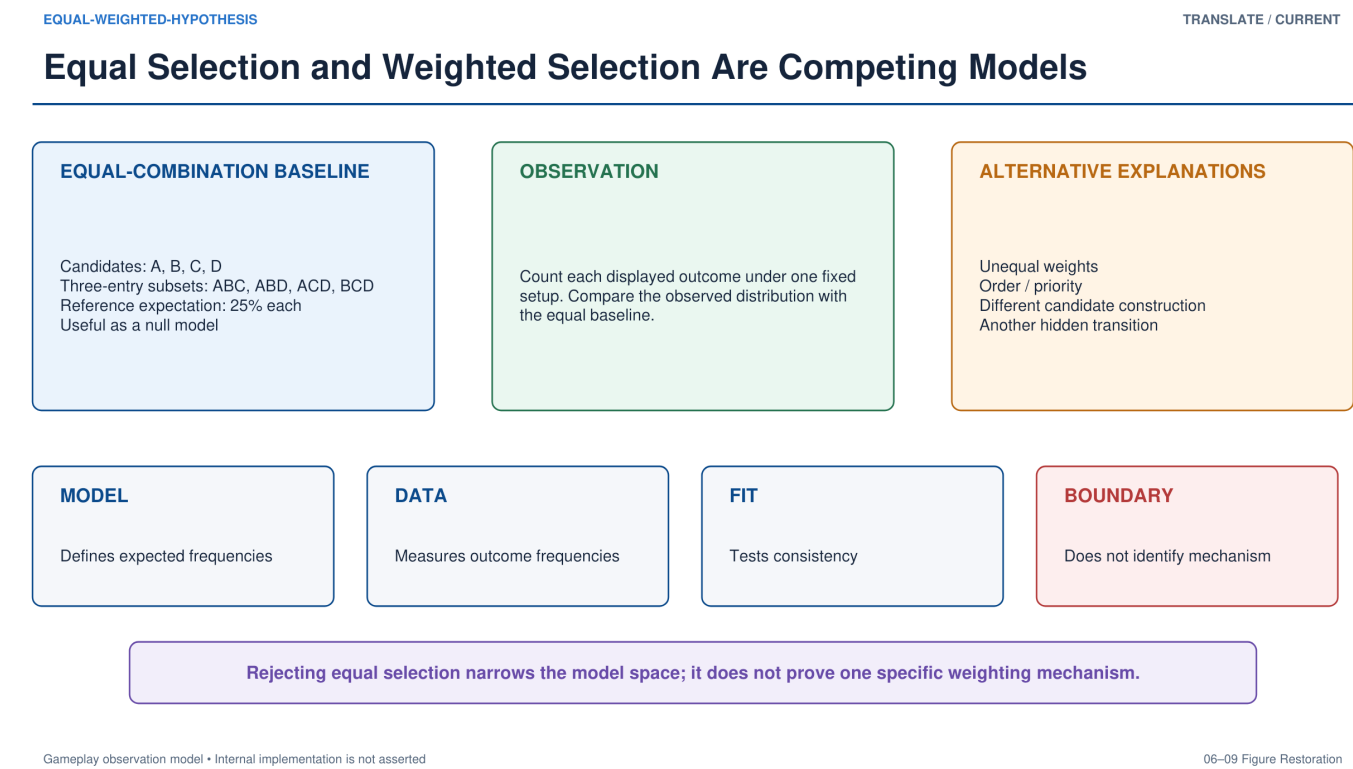
Glitter Augite 6-stack	100	58	58%	57.14%
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For each condition, a chi-square goodness-of-fit comparison detected no statistically significant departure from the stated reference probability ($p > 0.05$). Thus, although Chapter 07 found a material-condition-dependent position bias in a different Object X experiment, no collapse of the Candidate Count Model was detected in these four RF4SP Messhilite conditions. Their retained-target rates remained consistent with the same combination-space reference model proposed for the RF5 series.

A direct matched-stack material comparison also detected no significant difference between Love Crystal and Glitter Augite in this sample. At 3-stack, the observed rates were 29% and 28% (Pearson chi-square without continuity correction, p approximately 0.876). At 6-stack, they were 59% and 58% (p approximately 0.886). These comparisons do not prove that material identity never matters; they show only that no material-specific difference was detected between these two materials under the tested 100-trial conditions.

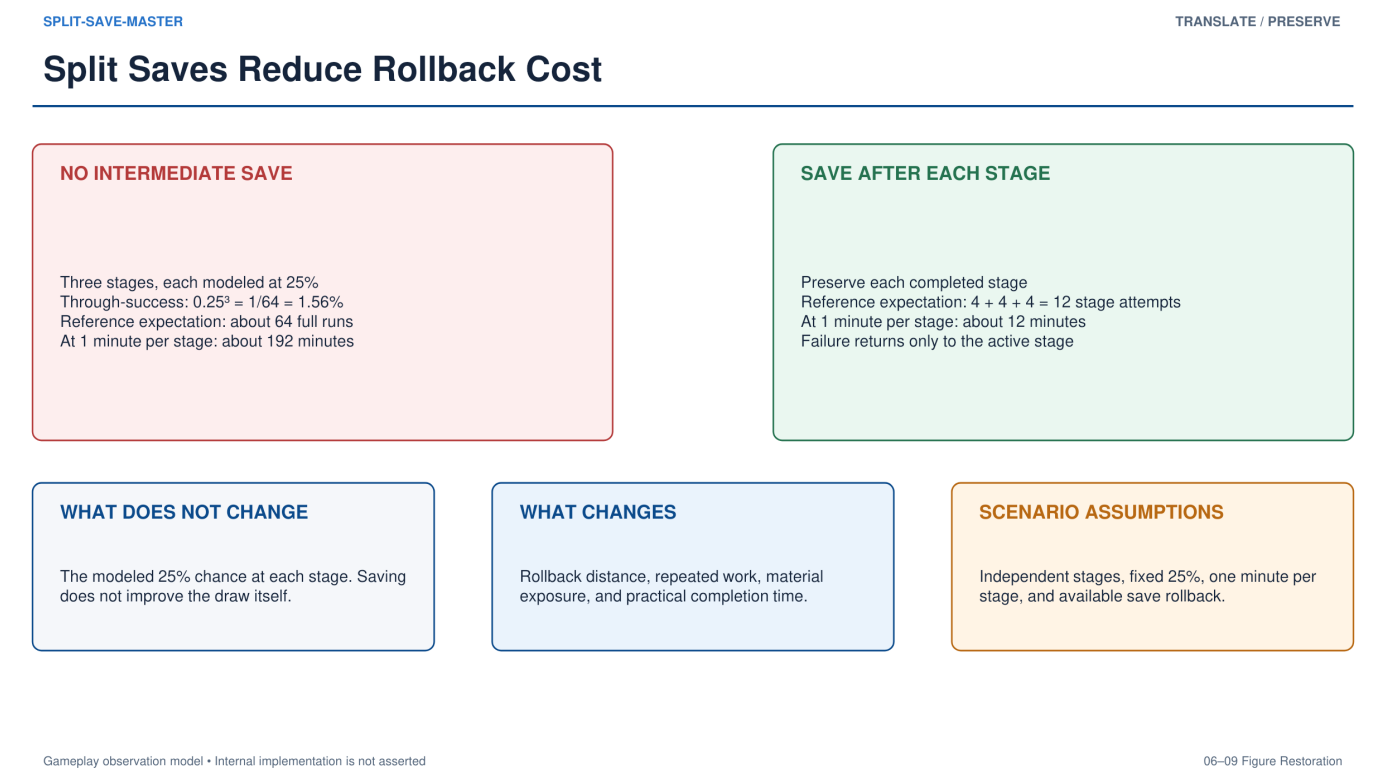
This result does not prove the model, equal selection, independence between subset selection and final permutation, or a shared internal mechanism between RF4SP and RF5. It means only that these four 100-trial observations did not significantly diverge from the condition-specific reference values at the available resolution. The internal relation between retained-subset formation and final ordering remains Unknown.

Small-sample exploratory comparison



Placement caption

Equal/weighted selection with process detail
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.



Placement caption

Split-save expected-attempt reduction model
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

RF4SP was also tested in the same general direction using five trials per condition for additional same-material and unrelated-material comparisons. These very small samples are exploratory only. They suggested the same practical direction, but cannot support a precise rate comparison or a uniformity claim.

The defensible cross-title interpretation is therefore:

In both RF4SP and RF5, current observations support same-material stacking as a useful practical direction for Messhilite inheritance involving Glitter Augite, Love Crystal, or another three-copy target.

RF5 provides the 4,000-trial primary series; RF4SP provides a separate 400-trial four-condition comparison plus smaller exploratory checks.

RF4SP and RF5 are not assumed to use the same internal process.

Chapter 07 also showed that RF4SP Object X position behavior was material-condition dependent: four elemental-material conditions were compatible with an equal-position approximation at 50 trials each, while the Mealy Apple condition strongly rejected it at 300 trials. That result prevents a general claim that all RF4SP order or selection behavior is uniform or non-uniform.

8-7. Actual-performance contamination and operation order

Messhilite in the arrange slots is not merely cosmetic

An additional RF5 comparison used Love Crystal effects to test whether Messhilite Ore occupying an arrange slot changed actual performance.

Arrange result	Observed drain rate	Approximate recovery at about 1,000 damage
Love Crystal x3	9%	About 90
Love Crystal x2 + Messhilite Ore	6%	About 60

The combat check contained only a little over ten attacks and is not a standalone statistical proof. However, the displayed 9% versus 6% difference and the approximately corresponding recovery amounts support the interpretation that Messhilite Ore displaced one Love Crystal effect under the tested condition. Therefore, excluding Messhilite Ore from the final three arrange slots has practical value: it helps preserve actual performance, not merely a preferred display.

Upgrade after inheritance

In an RF5 in-game check, properties added through the weapon's upgrade slots did not carry over to the inherited weapon under the tested condition. To avoid spending time optimizing properties that will be lost, the safer operational sequence is:

1. Complete Messhilite inheritance.
2. Confirm the final weapon and arrange result.
3. Apply upgrades to the completed final weapon.

This is an RF5 observation limited to the tested setup. It is not a universal claim about every transfer path or title.

8-8. Inherited weapon category and shield performance

An additional in-game observation indicates that Messhilite inheritance may preserve a weapon-category property from the **performance source**, rather than using only the visible category of the completed weapon, when shield performance is calculated.

Under the tested conditions:

- A shield-compatible performance source inherited into dual blades or fists allowed shield performance to be expressed, even though the completed weapon's visible category was dual blades or fists.
- In the reverse direction, using dual blades or fists as the performance source and inheriting them into a normally shield-compatible completed weapon did not produce the expected shield performance.

Performance-source category	Completed visible category	Observed shield result
Shield-compatible weapon	Dual blades or fists	Shield performance could be expressed
Dual blades or fists	Shield-compatible weapon	Shield performance was not expressed as expected

The directionality is consistent with the interpretation that the inherited performance source retains a category-dependent property relevant to shield calculation. It does not establish that every weapon-category-dependent mechanic follows the same rule.

Practical caution: when dual blades or fists are used as the performance source, verify shield behavior rather than assuming that the final weapon's visible category restores normal shield performance.

8-9. Practical workflow

1. Define success before crafting: specify the exact three-material arrange result and whether order matters.
2. Keep unrelated candidates out of the pool where possible.
3. For a three-copy target, choose the stack level according to cost and retry tolerance: 4-stack for efficiency or 5-stack for stability.
4. Save at a point that allows the failed randomization step to be repeated without rebuilding earlier stages.
5. Inspect both the arrange display and the actual resulting performance.
6. Finish weapon upgrades only after the final inheritance result is confirmed.

The optimal theoretical probability is not always the optimal player workflow. Material cost, reset time, judgment time, inventory, and tolerance for repetition all matter.

8-10. Interpretation after Chapters 06 and 07

The present evidence supports three separate statements:

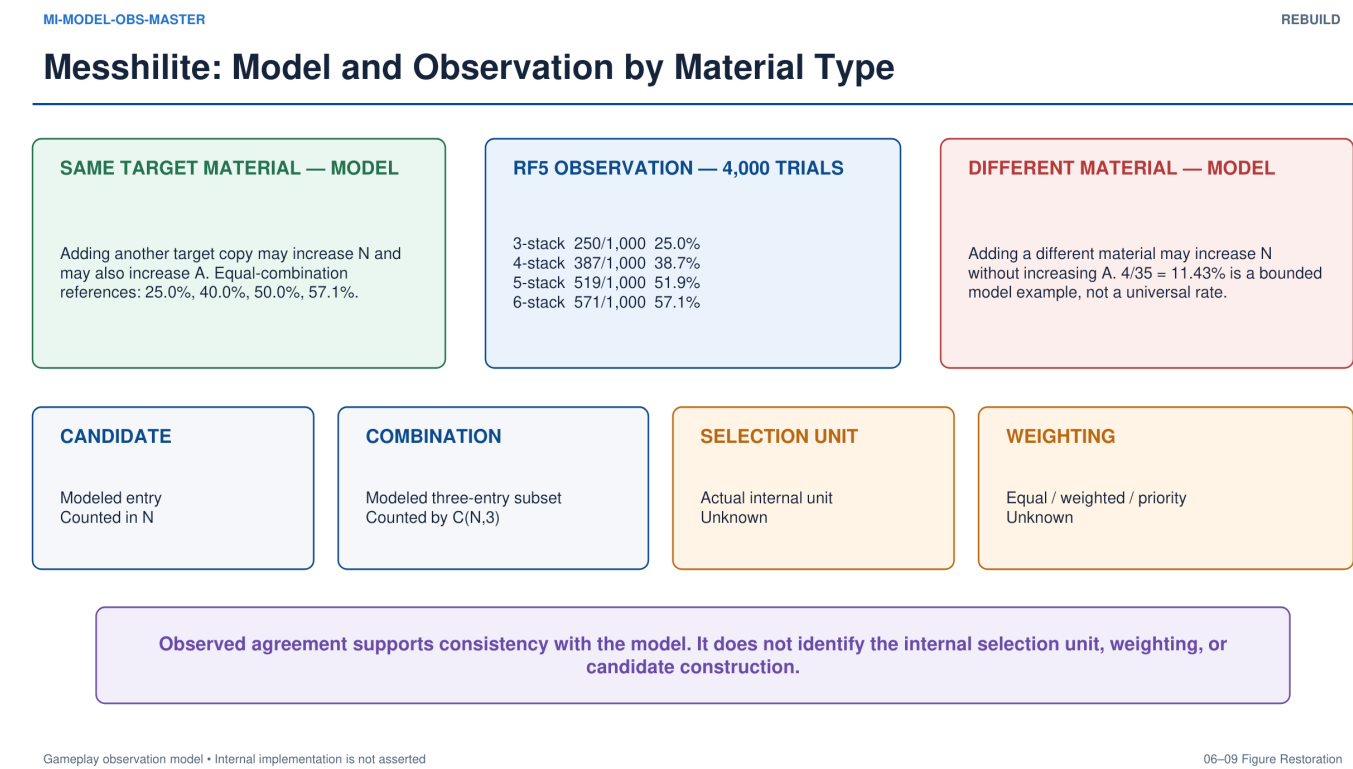
- 1. Combination space:** Increasing copies of the same target material increases the number of three-candidate subsets that satisfy the target condition.
- 2. Observed RF5 and RF4SP distributions:** The RF5 4,000-trial series and the post-permutation-problem RF4SP four-condition, 400-trial series were consistent with their condition-specific equal-selection reference values at the available resolution; model collapse was not detected under those Messhilite conditions.
- 3. Mechanism:** The internal candidate generation, weighting, selection, ordering, and transfer sequence remain Unknown.

The first statement is combinatorial. The second is empirical and condition-specific. The third is a boundary declaration.

The turnip-family subset-selection observation and the Mealy Apple Object X position observation do not falsify the RF5 Messhilite results. They show why the agreement observed here must not be generalized into a universal equal-randomness rule.

Figure 08-P22-img-017

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Analytically, the research can be organized as:

Candidate collection -> candidate set -> combination space -> retained subset -> final order -> observed outcome

This is an observation framework, not a claim that the game implements six separate processing stages.

8-11. Reproducibility and future collaborative validation

The current package demonstrates that an individual player can collect a large repeated-observation dataset, but it also exposes the cost of doing so. A community-scale extension could use a shared form and spreadsheet to collect:

- title and platform;
- patch/version information where known;
- source and destination equipment;
- exact candidate materials and stack counts;
- success definition;
- observed arrange result and order;
- actual-performance check;
- save/reload procedure;
- trial identifier and contributor identifier.

Shared collection would require a written protocol, duplicate-control rules, validation of judgment criteria, and preservation of raw records. More data without consistent conditions would not necessarily improve the evidence.

The deeper purpose of this chapter is therefore not only the Candidate Count Model. It is to demonstrate a reusable research process:

Define what must be observed, separate competing conditions, state the success criterion, collect repeated observations, compare them with a replaceable model, and return to Reality when the model and observations diverge.

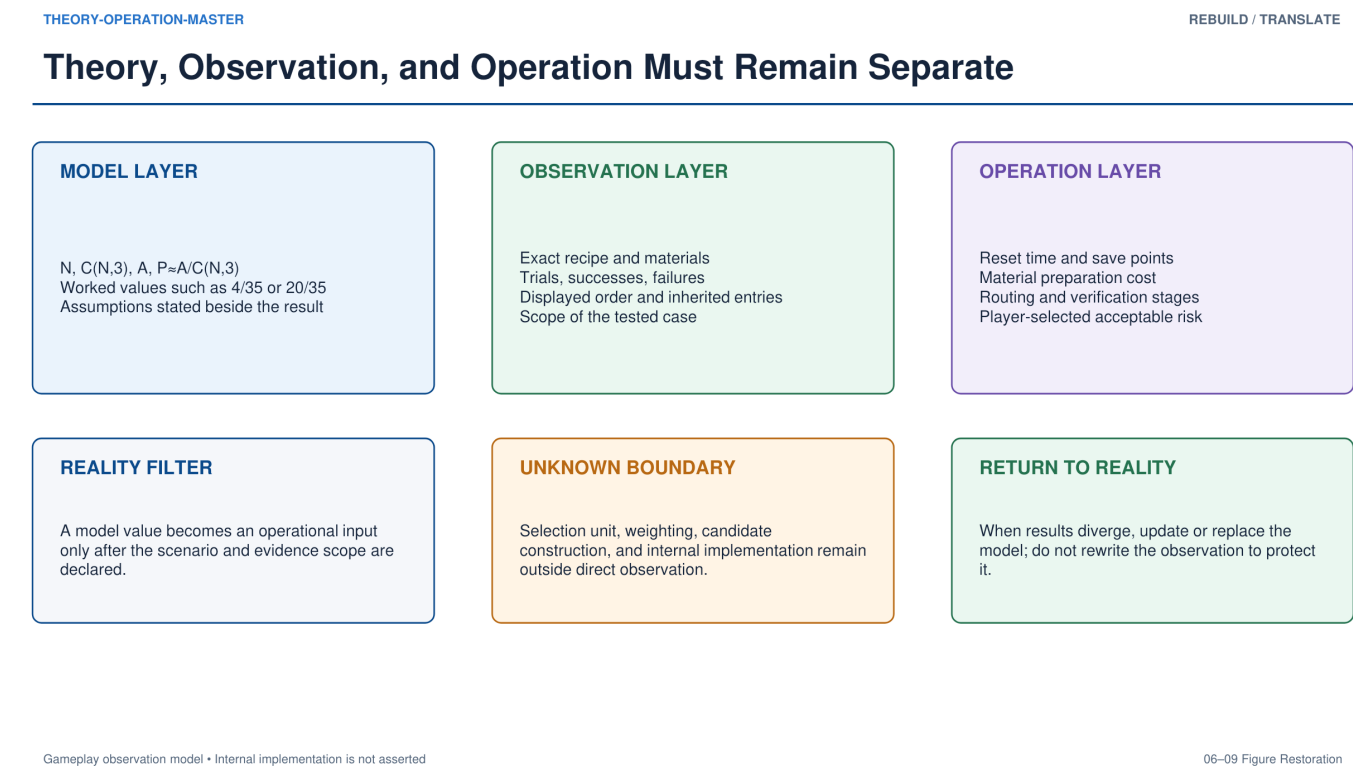
If that process helps another player formulate and test a new question, it is a more durable outcome than any single probability estimate.

Evidence boundary and status

- **RF5 4,000-trial same-material series:** observed and aggregated.
- **RF5 observed rates near the equal-selection reference model:** supported under the tested conditions.
- **4-stack efficiency / 5-stack stability recommendation:** practical interpretation of the observed marginal gains.
- **RF4SP four-condition comparison: 100 trials per condition, 400 total; no significant departure from the stated reference values was detected ($p > 0.05$), and no matched-stack difference was detected between Love Crystal and Glitter Augite. Additional five-trial comparisons remain exploratory.**
- **Messhilite-related performance contamination:** observed under the Love Crystal comparison condition.
- **Upgrade-after-inheritance order:** observed in RF5 under the tested condition.
- **Performance-source weapon category and shield behavior:** directional effect observed under the tested Messhilite conditions; broader category-dependent mechanics remain unestablished.
- **Universal equal selection, universal equal permutation, or shared RF4SP/RF5 internal implementation:** not established.
- **Selection and order mechanism:** Unknown.

Figure 08-P23-img-018

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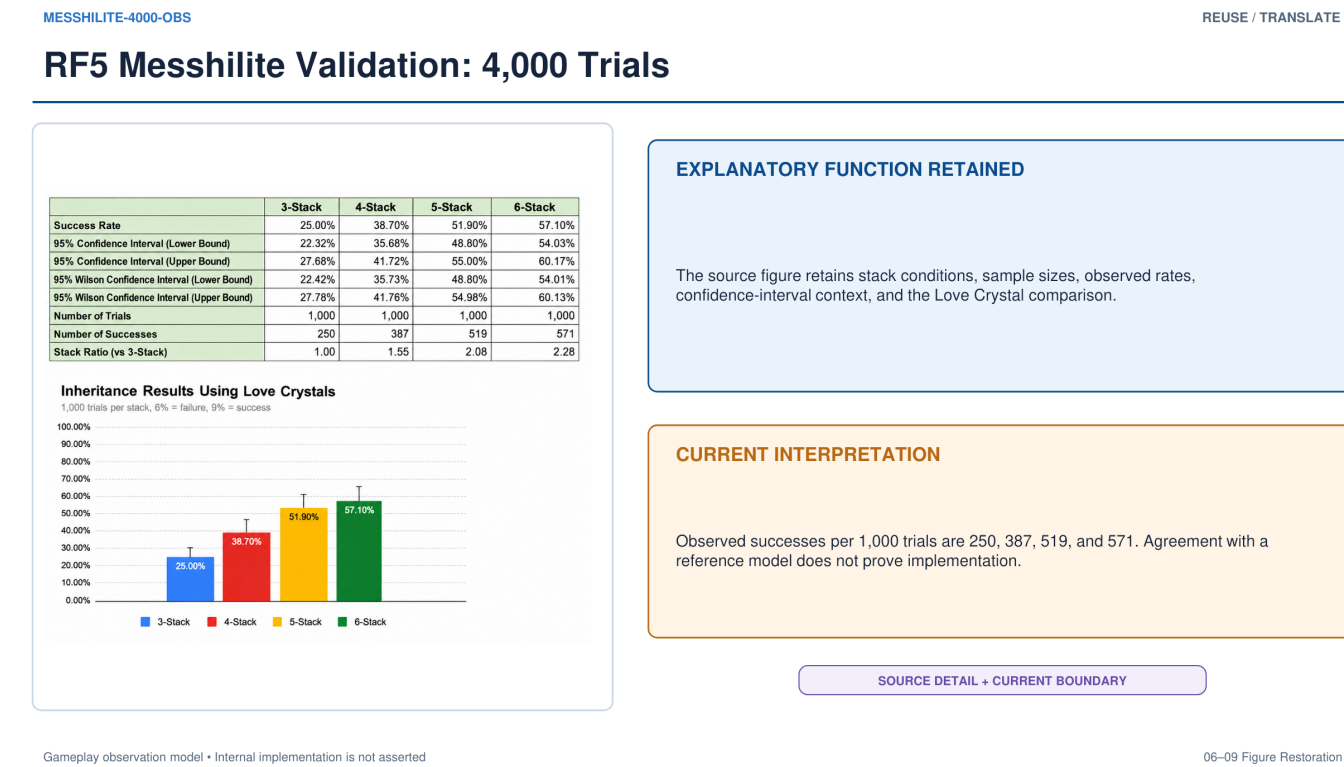


Placement caption

Theory versus operational success model
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Figure 08-P25-img-019

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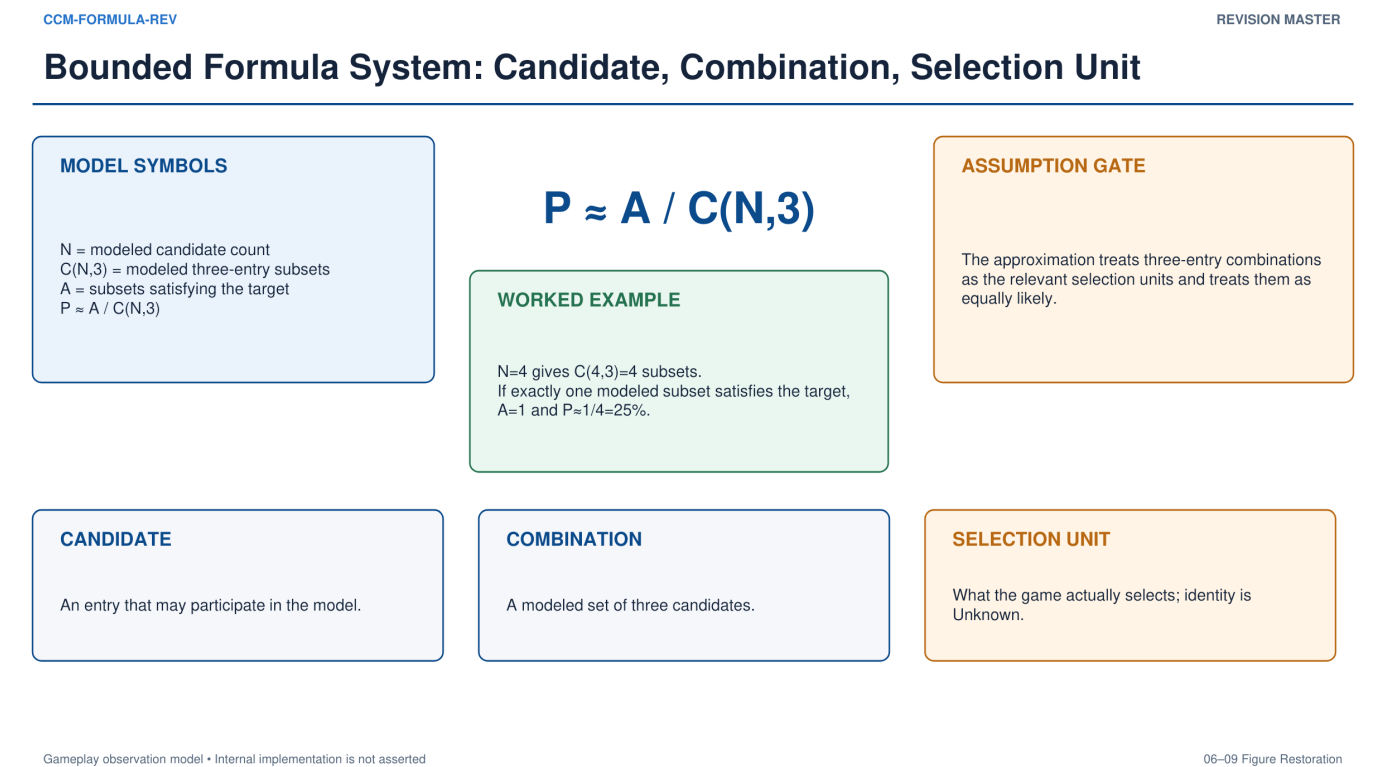
Love Crystal absorption comparison observation
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Supplementary research package

The accompanying package contains the validation methodology, Wilson confidence-interval derivation, aggregated results, four condition-level records, integrated raw CSV, and a note on validation efficiency.

Figure 08-P28-img-020

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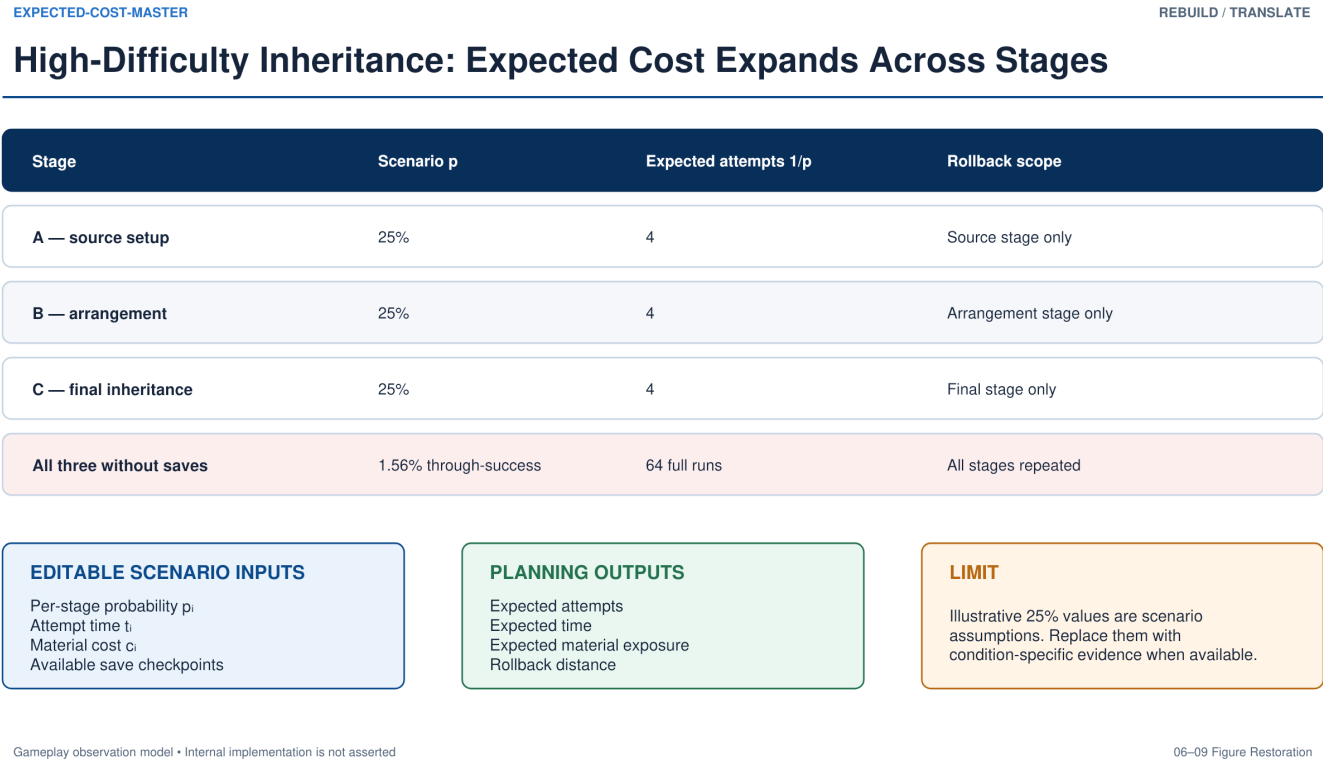


Placement caption

High-difficulty inheritance research overview
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Figure 08-P30-img-021

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Placement caption

Expected-value explosion model
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.